

# WEEK 7 PROJECT REPORT

## DATA CLEANING & TRANSFORMATION

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**Project Title:** Student Performance – Data Cleaning & Transformation

**Tool Used:** Microsoft Power BI Desktop

**Dataset:** Students Performance Dataset

## INTRODUCTION

The objective of this Week 7 Power BI task was to clean, transform, and prepare the **Student Performance in Exams** dataset for analysis and visualization. The workflow was completed using **Power BI Desktop** and **Power Query Editor**. The dataset was inspected for data quality issues, categorical attributes were standardized, calculated columns were created, student performance was classified, and subject-wise data was reshaped using unpivoting.

## IMPORTING THE DATASET

The Student Performance in Exams dataset was imported into **Power BI Desktop** using the **Get Data** option. The appropriate source format, such as Text/CSV or Excel, was selected and the dataset was loaded into Power Query Editor using **Transform Data**.

### StudentsPerformance.csv

File Origin

1252: Western European (Windows)

Delimiter

Comma

Data Type Detection

Based on first 200 rows

| gender | race/ethnicity | parental level of education | lunch        | test preparation course | math score | reading score | writing score |
|--------|----------------|-----------------------------|--------------|-------------------------|------------|---------------|---------------|
| female | group B        | bachelor's degree           | standard     | none                    | 72         | 72            | 74            |
| female | group C        | some college                | standard     | completed               | 69         | 90            | 88            |
| female | group B        | master's degree             | standard     | none                    | 90         | 95            | 93            |
| male   | group A        | associate's degree          | free/reduced | none                    | 47         | 57            | 44            |
| male   | group C        | some college                | standard     | none                    | 76         | 78            | 75            |
| female | group B        | associate's degree          | standard     | none                    | 71         | 83            | 78            |
| female | group B        | some college                | standard     | completed               | 88         | 95            | 92            |
| male   | group B        | some college                | free/reduced | none                    | 40         | 43            | 39            |
| male   | group D        | high school                 | free/reduced | completed               | 64         | 64            | 67            |
| female | group B        | high school                 | free/reduced | none                    | 38         | 60            | 50            |
| male   | group C        | associate's degree          | standard     | none                    | 58         | 54            | 52            |
| male   | group D        | associate's degree          | standard     | none                    | 40         | 52            | 43            |
| female | group B        | high school                 | standard     | none                    | 65         | 81            | 73            |
| male   | group A        | some college                | standard     | completed               | 78         | 72            | 70            |
| female | group A        | master's degree             | standard     | none                    | 50         | 53            | 58            |
| female | group C        | some high school            | standard     | none                    | 69         | 75            | 78            |
| male   | group C        | high school                 | standard     | none                    | 88         | 89            | 86            |
| female | group B        | some high school            | free/reduced | none                    | 18         | 32            | 28            |
| male   | group C        | master's degree             | free/reduced | completed               | 46         | 42            | 46            |
| female | group C        | associate's degree          | free/reduced | none                    | 54         | 58            | 61            |

The data in the preview has been truncated due to size limits.

Extract Table Using Examples

Load

Transform Data

Cancel

## EXPLORING AND INSPECTING THE DATA

The dataset was examined to understand its structure, columns, rows, and data types. The available attributes included student demographic information and examination scores.

The following types of attributes were identified:

- **Categorical attributes:** Gender, Race/Ethnicity, Parental Education, Lunch, and Test Preparation Course.
- **Numerical attributes:** Math Score, Reading Score, and Writing Score.

The columns were checked for appropriate data types, missing values, invalid values, and duplicate records. This inspection helped identify the transformations required before analysis.

| = Table.RenameColumns("#Reordered Columns",{{"Index", "StudentID"}, {"gender", "Gender"}, {"race/ethnicity", "Race/Ethnicity"}, {"parental |                  |                                                             |                  |                                                             |                  |                                                             |  |  |  |
|--------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------|------------------|-------------------------------------------------------------|------------------|-------------------------------------------------------------|--|--|--|
|                                                                                                                                            | StudentID        | Gender                                                      | Race/Ethnicity   | Parental level of Education                                 | Lunch            | Test Preparation Course                                     |  |  |  |
| <div><div>Valid</div><div>Error</div><div>Empty</div></div>                                                                                | 100%<br>0%<br>0% | <div><div>Valid</div><div>Error</div><div>Empty</div></div> | 100%<br>0%<br>0% | <div><div>Valid</div><div>Error</div><div>Empty</div></div> | 100%<br>0%<br>0% | <div><div>Valid</div><div>Error</div><div>Empty</div></div> |  |  |  |
| 1                                                                                                                                          |                  | 1 female                                                    | group B          | bachelor's degree                                           | standard         | none                                                        |  |  |  |
| 2                                                                                                                                          |                  | 2 female                                                    | group C          | some college                                                | standard         | completed                                                   |  |  |  |
| 3                                                                                                                                          |                  | 3 female                                                    | group B          | master's degree                                             | standard         | none                                                        |  |  |  |
| 4                                                                                                                                          |                  | 4 male                                                      | group A          | associate's degree                                          | free/reduced     | none                                                        |  |  |  |
| 5                                                                                                                                          |                  | 5 male                                                      | group C          | some college                                                | standard         | none                                                        |  |  |  |
| 6                                                                                                                                          |                  | 6 female                                                    | group B          | associate's degree                                          | standard         | none                                                        |  |  |  |
| 7                                                                                                                                          |                  | 7 female                                                    | group B          | some college                                                | standard         | completed                                                   |  |  |  |
| 8                                                                                                                                          |                  | 8 male                                                      | group B          | some college                                                | free/reduced     | none                                                        |  |  |  |
| 9                                                                                                                                          |                  | 9 male                                                      | group D          | high school                                                 | free/reduced     | completed                                                   |  |  |  |
| 10                                                                                                                                         |                  | 10 female                                                   | group B          | high school                                                 | free/reduced     | none                                                        |  |  |  |
| 11                                                                                                                                         |                  | 11 male                                                     | group C          | associate's degree                                          | standard         | none                                                        |  |  |  |
| 12                                                                                                                                         |                  | 12 male                                                     | group D          | associate's degree                                          | standard         | none                                                        |  |  |  |
| 13                                                                                                                                         |                  | 13 female                                                   | group B          | high school                                                 | standard         | none                                                        |  |  |  |
| 14                                                                                                                                         |                  | 14 male                                                     | group A          | some college                                                | standard         | completed                                                   |  |  |  |
| 15                                                                                                                                         |                  | 15 female                                                   | group A          | master's degree                                             | standard         | none                                                        |  |  |  |
| 16                                                                                                                                         |                  | 16 female                                                   | group C          | some high school                                            | standard         | none                                                        |  |  |  |
| 17                                                                                                                                         |                  | 17 male                                                     | group C          | high school                                                 | standard         | none                                                        |  |  |  |
| 18                                                                                                                                         |                  | 18 female                                                   | group B          | some high school                                            | free/reduced     | none                                                        |  |  |  |
| 19                                                                                                                                         |                  | 19 male                                                     | group C          | master's degree                                             | free/reduced     | completed                                                   |  |  |  |
| 20                                                                                                                                         |                  | 20 female                                                   | group C          | associate's degree                                          | free/reduced     | none                                                        |  |  |  |
| 21                                                                                                                                         |                  | 21 male                                                     | group D          | high school                                                 | standard         | none                                                        |  |  |  |
| 22                                                                                                                                         |                  | 22 female                                                   | group B          | some college                                                | free/reduced     | completed                                                   |  |  |  |
| 23                                                                                                                                         |                  | 23 male                                                     | group D          | some college                                                | standard         | none                                                        |  |  |  |
| 24                                                                                                                                         |                  | 24 female                                                   | group C          | some high school                                            | standard         | none                                                        |  |  |  |
| 25                                                                                                                                         |                  | 25 male                                                     | group D          | bachelor's degree                                           | free/reduced     | completed                                                   |  |  |  |

## CLEANING THE DATASET

The dataset was cleaned in Power Query to improve its quality and consistency.

The following cleaning operations were performed:

- Duplicate records were removed where appropriate.
- Missing or invalid values were checked and handled.
- Appropriate data types were assigned to the columns.
- Inconsistent categorical values were standardized.
- Column names were reviewed and renamed where necessary for better clarity.

These steps ensured that the dataset was reliable and suitable for further transformation and analysis.

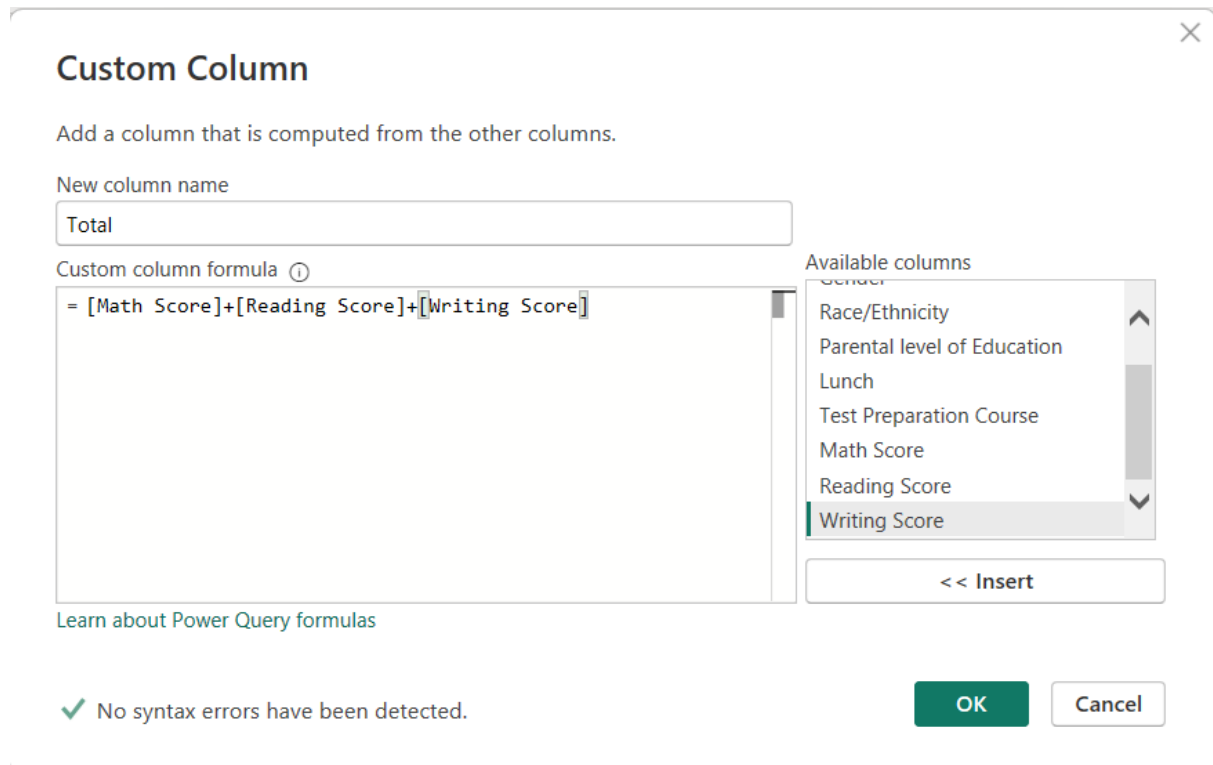
## CREATING CALCULATED COLUMNS IN POWER QUERY

### 1. Creating the Total Score

A Total column was created in Power Query using a Custom Column. The total score was calculated by adding the student's Math, Reading, and Writing scores.

The formula used was: **[Math Score] + [Reading Score] + [Writing Score]**

This provides the combined score of the student across all three subjects.



**Custom Column**

Add a column that is computed from the other columns.

New column name

Total

Custom column formula ⓘ

= [Math Score]+[Reading Score]+[Writing Score]

Available columns

- Gender
- Race/Ethnicity
- Parental level of Education
- Lunch
- Test Preparation Course
- Math Score
- Reading Score
- Writing Score

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK Cancel

### 2. Creating the Average Score

An Average column was then created using the Total Score.

The formula used was: **[Total] / 3**

This calculates the student's average performance across Math, Reading, and Writing.

Custom Column

×

Add a column that is computed from the other columns.

New column name

Average

Custom column formula ⓘ

= [Total]/3

Available columns

Parental level of Education

Lunch

Test Preparation Course

Math Score

Reading Score

Writing Score

Total

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK

Cancel

## CREATING THE PERFORMANCE GRADE

A Grade column was created using Power Query's Conditional Column feature. The student's Average Score was used to classify performance into different grades.

The conditions used in the completed workflow were:

| Average Score | Grade   |
|---------------|---------|
| ≥ 90          | Grade A |
| ≥ 75          | Grade B |
| ≥ 50          | Grade C |
| ≥ 40          | Grade D |
| Below 40      | Fail    |

This classification provides a simple way to identify the performance level of each student.

×

### Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name  
Grade

|         | Column Name | Operator              | Value | Output  |
|---------|-------------|-----------------------|-------|---------|
| If      | Average     | is greater than or... | 90    | Grade A |
| Else If | Average     | is greater than or... | 75    | Grade B |
| Else If | Average     | is greater than or... | 50    | Grade C |
| Else If | Average     | is greater than or... | 40    | Grade D |

Add Clause

Else  
Fail

OK

Cancel

## CREATING PASS/FAIL CLASSIFICATION

A Pass/Fail column was created using a Conditional Column.

The criterion used was:

- Average > 40 → Pass
- Otherwise → Fail

This classification allows students to be easily separated into passing and failing groups.

×

### Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name  
Pass/Fail

|    | Column Name | Operator        | Value | Output |
|----|-------------|-----------------|-------|--------|
| If | Average     | is greater than | 40    | Pass   |

Add Clause

Else  
Fail

OK

Cancel

## RESHAPING SUBJECT DATA USING UNPIVOT

To make the dataset more suitable for subject-wise analysis, the three subject score columns were transformed using the **Unpivot Columns** operation.

The following columns were selected:

- Math Score
- Reading Score
- Writing Score

After unpivoting, the resulting columns were renamed appropriately:

- **Subject**
- **Score**

This transformation converted the subject columns into rows, allowing the performance of students to be analyzed consistently across different subjects.

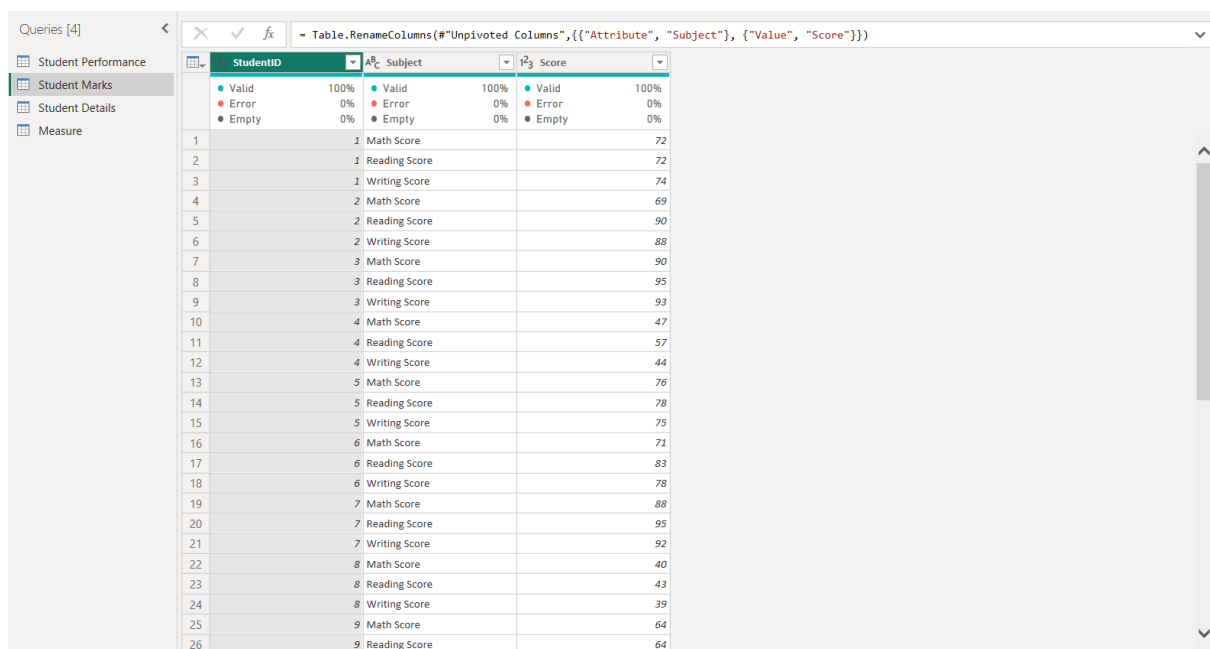


Table: Table.RenameColumns(#"Unpivoted Columns",{{"Attribute", "Subject"}, {"Value", "Score"}})

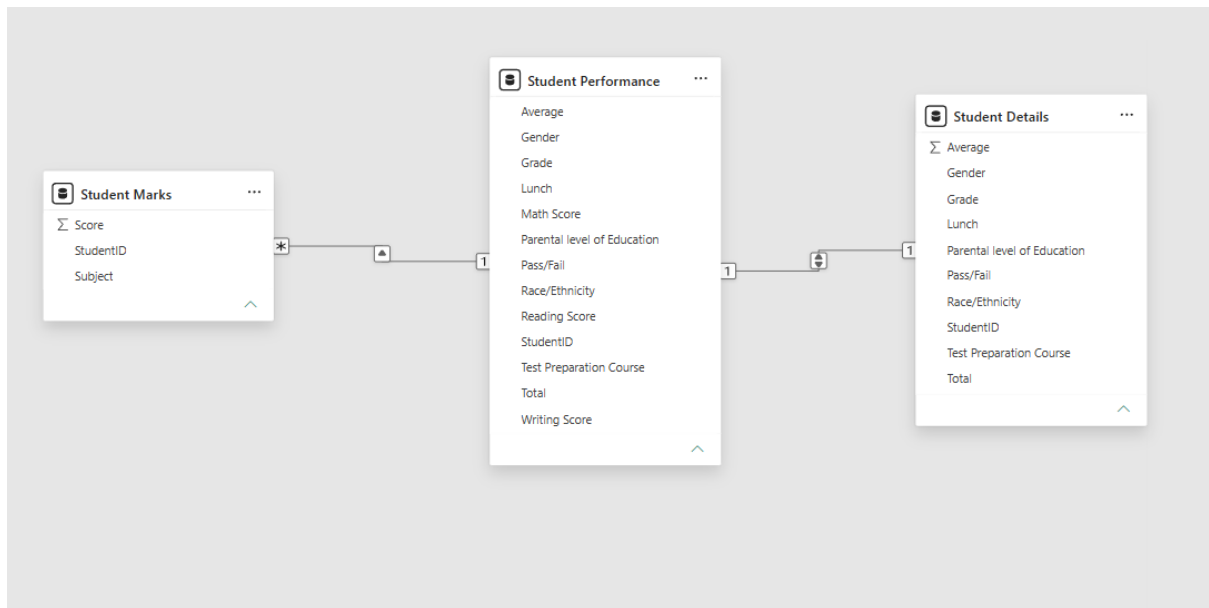
| StudentID | Subject       | Score |
|-----------|---------------|-------|
| 1         | Math Score    | 72    |
| 2         | Reading Score | 72    |
| 3         | Writing Score | 74    |
| 4         | Math Score    | 69    |
| 5         | Reading Score | 90    |
| 6         | Writing Score | 88    |
| 7         | Math Score    | 90    |
| 8         | Reading Score | 95    |
| 9         | Writing Score | 93    |
| 10        | Math Score    | 47    |
| 11        | Reading Score | 57    |
| 12        | Writing Score | 44    |
| 13        | Math Score    | 76    |
| 14        | Reading Score | 78    |
| 15        | Writing Score | 75    |
| 16        | Math Score    | 71    |
| 17        | Reading Score | 83    |
| 18        | Writing Score | 78    |
| 19        | Math Score    | 88    |
| 20        | Reading Score | 95    |
| 21        | Writing Score | 92    |
| 22        | Math Score    | 40    |
| 23        | Reading Score | 43    |
| 24        | Writing Score | 39    |
| 25        | Math Score    | 64    |
| 26        | Reading Score | 64    |

## PREPARING THE DATA MODEL

After completing the transformations, the resulting table was reviewed to ensure that the structure was appropriate for analysis.

Unnecessary columns were removed where required, and the transformed fields were checked for appropriate data types and consistency. The resulting structure was prepared for use within the Power BI data model.

The model was reviewed to ensure that the required fields were available for calculations, filtering, and visualization.



## CREATING DAX MEASURES

Basic DAX measures were created to calculate important performance indicators.

### 1. Average Math Score

This measure calculates the average score obtained by students in Mathematics.

### 2. Average Reading Score

This measure calculates the average score obtained by students in Reading.

### 3. Average Writing Score

This measure calculates the average score obtained by students in Writing.

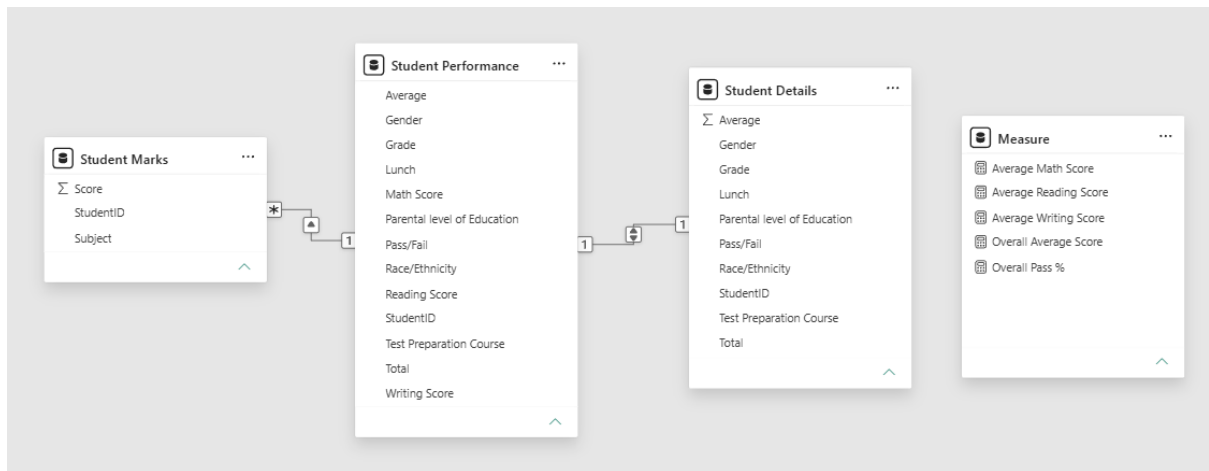
### 4. Overall Average Score

This measure represents the overall average performance of students.

### 5. Overall Pass %

This measure calculates the percentage of students who passed according to the defined pass criterion.

These measures provide key indicators that can later be used in Power BI cards, charts, and other visualizations.



## CONCLUSION

The Week 7 task successfully demonstrated the process of preparing a real-world student performance dataset for analysis using Power BI. Data cleaning techniques were applied to improve data quality, categorical fields were standardized, calculated columns were created, students were classified based on performance and pass/fail status, and subject data was reshaped using unpivoting.

The creation of DAX measures further enhanced the analytical capabilities of the dataset. Overall, the completed workflow produced a structured and analysis-ready dataset that can be effectively used for student performance visualization and dashboard development.